

Forecasting Hidden Simulators

Learning 10 systems from limited data

0.34 → **0.77**

How I read the problem

The constraint

What it meant for my strategy

Only 2,000 steps of data



Every step has to answer a question

Predict 4,000 steps blind



Need real structure, not curve fits

3 uploads a day



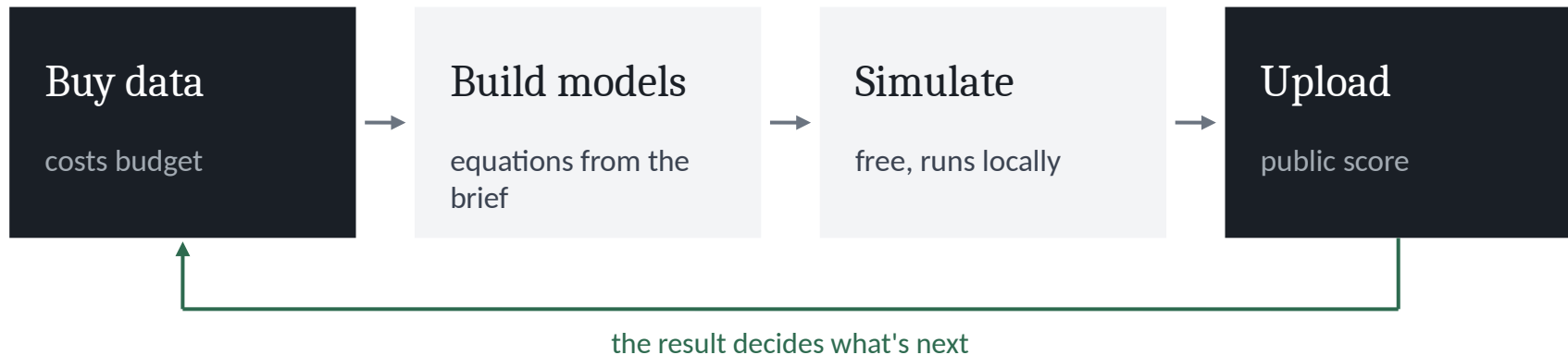
Every upload is an experiment

Score rewards being close



Don't hedge toward the average

My process: one loop, every day



Key idea

- Dark boxes are scarce: real data and uploads
- Light boxes are free: test as much as I want locally
- So: do the thinking for free, spend the scarce stuff carefully

Inside one model

1 Read the brief

pick stocks, delays, caps and the 3 named mechanisms

2 Write the ODE

≤ 8 states, ≤ 12 fitted parameters, RK4 integration

3 Fit it

robust (Cauchy) least squares, 10 random starts

4 Validate it

leave-one-run-out, scored at the organizers' σ

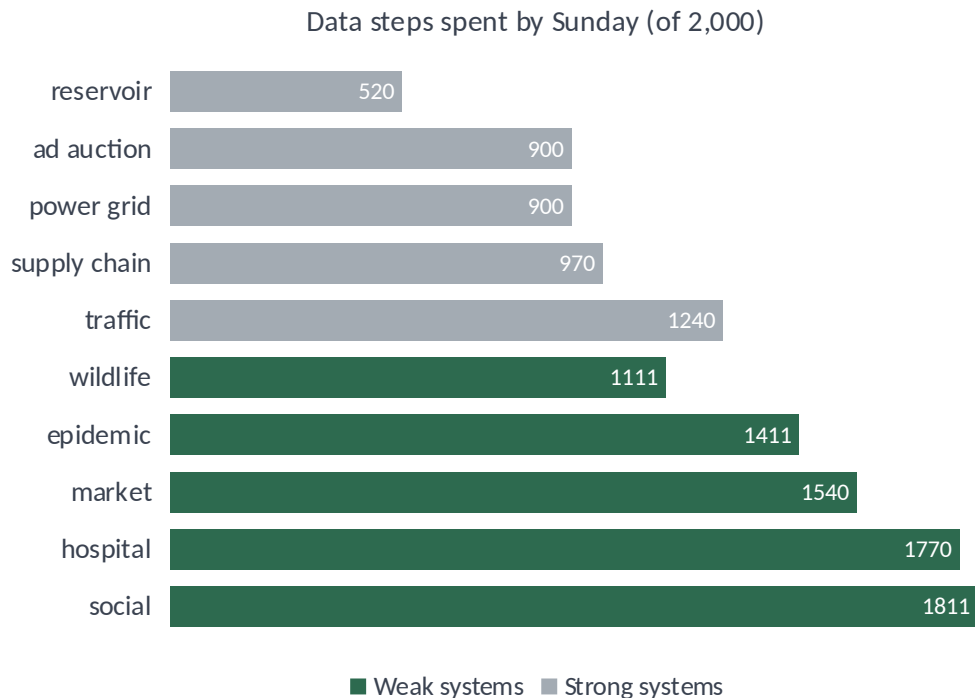
Example: reservoir

$$\frac{dL}{dt} = \text{inflow}(t) - c \cdot (L/500)^p - \text{seep}$$
$$L \leq 941 \quad (\text{hard cap})$$
$$\text{inflow}(t) = 11.43 + 2.15 \cdot \sin(2\pi t / 67.8)$$

outflow = delivery + spill when full
quality: first-order in aeration

Every term comes from a line in the brief. Only the constants are fitted.

Strong and weak systems got different playbooks



Strong (0.78–0.89)

- One structure clearly fits
- Refine details: delays, caps, bins
- Keep data in reserve

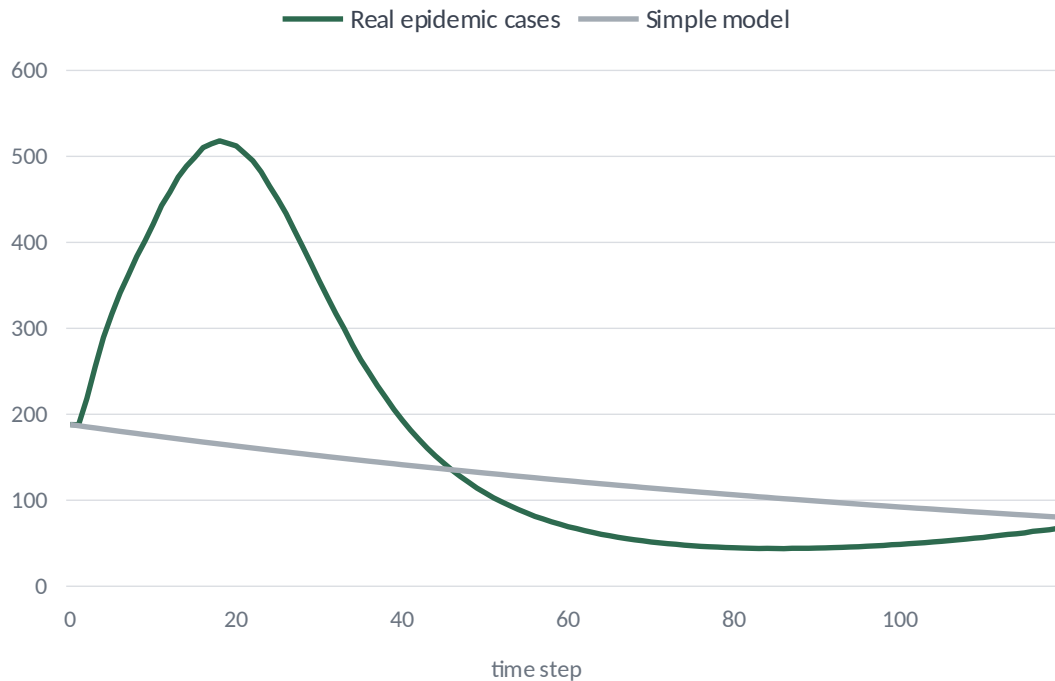
Weak (0.63–0.75)

- Scored 4–12× stricter
- Several structures fit equally
- Fit all 3 mechanism pairs
- Buy data where models disagree
- Ship a median of models if no clear winner

What each final model captures

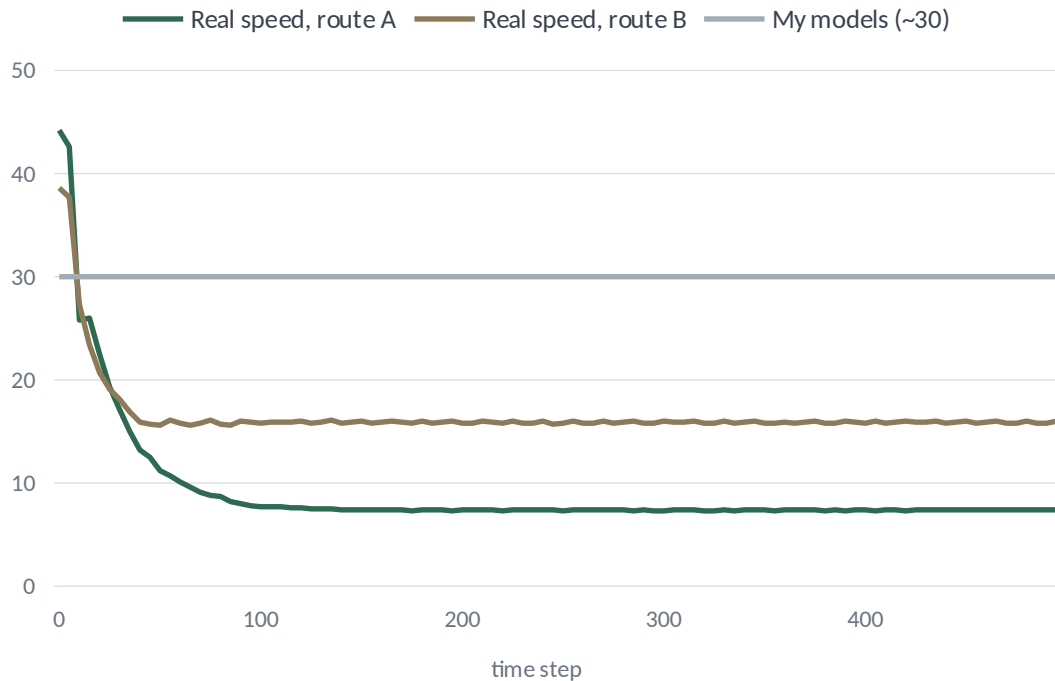
System	Score	Model
reservoir	0.89	seasonal inflow wave, groundwater inflow, slow water-quality stock
ad auction	0.88	audience tracked in 10 targeting-breadth bins
traffic	0.85	7-tick travel delay; lane closure slows and shrinks route B
supply chain	0.82	congestion + commitment; 3-tick pure delay
power grid	0.78	per-observable mix; thermostat-style load rebound
wildlife	0.75	median of 5 predator-prey models
hospital	0.73	mean of two queue models; exactly 2 ticks in service
epidemic	0.72	SIRS + hospital stock + 2-tick reporting delay
social	0.65	two-community diffusion with short memory
market	0.63	dealer inventory model with a freeze switch (hysteresis)

1. Structure over flexibility



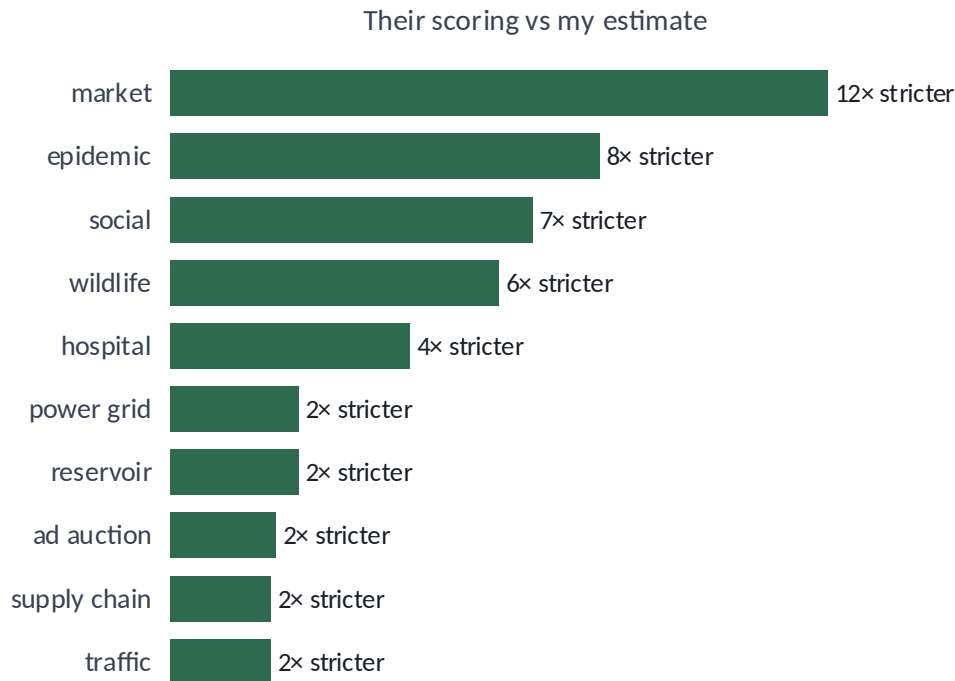
- Linear models couldn't fit their own training data
- Data shapes: waves, rebounds, caps, delays
- Grey-box ODE per system from the brief
- Small enough for the data to pin down

2. Spend data like an experiment



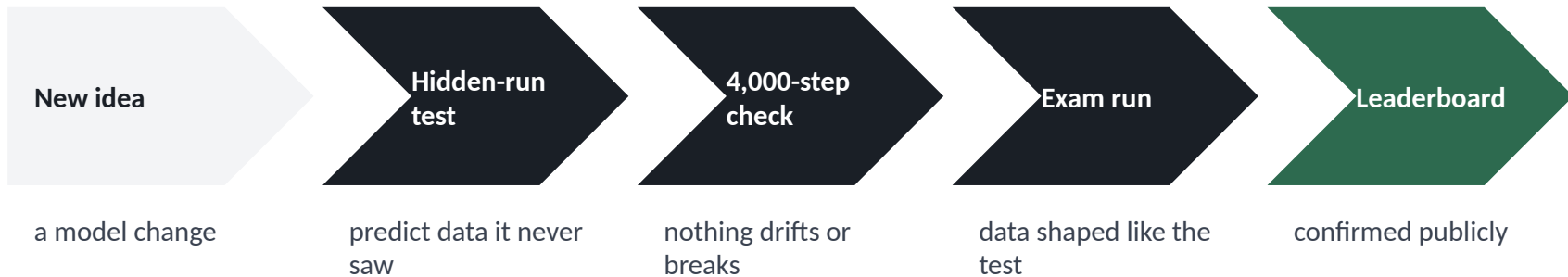
- Probe first: 120-step hold, measure time constants
- Size later runs to 3–5× the slowest time constant
- Committee of validated models on test-shaped schedules
- Buy where they disagree most (in σ units)

3. Use the leaderboard as a lab



- One change per system per upload: 10 A/B tests each
- Replayed every upload on my own runs
- Solved $\sigma_{\text{org}} = k \cdot \sigma_{\text{mine}}$ per system
- Refit everything at their σ

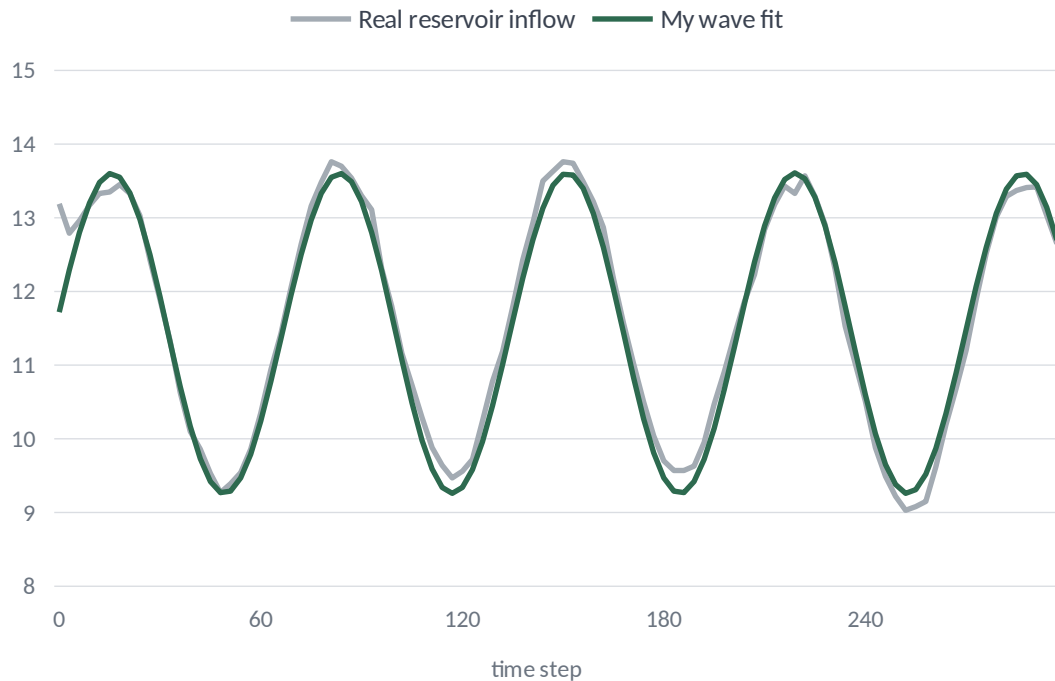
4. Every change had to earn its place



My rules for shipping

- Has to win across most test runs, not one lucky run
- The hardest test case can't get worse
- Final entry = the best leaderboard-confirmed version of each system

5. Look for how it was built



- Hunch: these are written code, not real physics
- Looked for exact repeats and hard switches
- Reservoir inflow: a fixed wave, not noise
- Market freeze has memory, hospital takes exactly 2 ticks

Summary

- Grey-box ODEs built from each brief, not black-box fits
- Data bought where models disagreed or had never looked
- Uploads run as A/B tests, refit at the organizers' scale
- Only changes that held up across runs got shipped

0.34 → **0.77** Questions?